Customer Segmentation

of AllLife Bank Credit-Card Customers



Share of customers across the three segments

Unsupervised learning project : K-Means & Hierarchical Clustering

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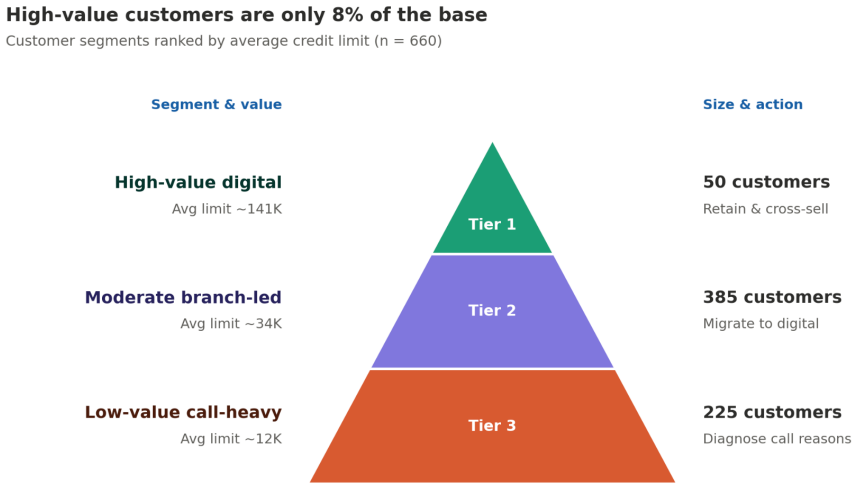
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Executive Summary

Analysis of **660 credit-card customers** using **K-Means and Hierarchical Clustering** to identify meaningful customer segments based on credit profile and interaction behaviour.

Three distinct customer segments identified:

- **High-value digital-first - 50 customers (~8%)**: very high credit limits, more cards, very high online usage, and minimal branch or phone interaction.
- **Moderate-value branch-oriented - 385 customers (~58%)**: moderate credit limits and card ownership, with a clear preference for branch visits and relatively low online and phone engagement.
- **Low-value call-heavy - 225 customers (~34%)**: lower credit limits, fewer cards, high call activity, and moderate online engagement.



Recommended segmentation approach:

K-Means with k = 3. Although Hierarchical Clustering achieved a higher silhouette score, K-Means identified the same broad segments while being substantially faster and more scalable for large banking datasets.

Top three recommendations:

1. Target **high-value digital-first customers** with premium cards, rewards, and personalized digital offers.
2. Encourage **moderate-value branch-oriented customers** toward gradual digital adoption while maintaining strong branch and hybrid support.
3. Improve service efficiency for **low-value call-heavy customers** through stronger call-center support and self-service options for common issues.

1. Problem framing & Objectives

Business Objective

AllLife Bank wants to deepen relationships with its credit-card customers. Today the marketing team has limited insight into how customers differ in value and in the way they interact with the bank, which makes it hard to target service, retention and growth efforts efficiently.

The objective of this analysis is to identify different segments in the existing customers, based on their spending patterns characterized by their credit limit as well as past interaction with the bank, using clustering algorithms, and provide recommendations to the bank on how to better market to and service these customers.

Data Science Problem

Problem type: Unsupervised Learning - Clustering

Algorithms used: KMeans, Hierarchical Clustering

Evaluation Approach

Algorithm	Evaluation Measure	Purpose
KMeans	Inertia / Elbow Method	Measures within-cluster compactness and helps identify a suitable number of clusters
	Silhouette Score	Measures how well observations are separated from neighbouring clusters
	Silhouette Diagram	Examines cluster-wise separation and identifies observations with weak or negative silhouette values
Hierarchical Clustering	Dendrogram	Examines the hierarchical structure and helps identify a suitable number of clusters
	Cophenetic Correlation	Measures how well the dendrogram preserves the pairwise distances in the original data
	Silhouette Score	Evaluates the separation and cohesion of the final cluster solution
	Silhouette Diagram	Examines the quality of assignments within individual clusters

Silhouette score will be used as the primary common metric for comparing final clustering solutions, while algorithm-specific diagnostics will support model and cluster-count selection.

2. About the Data

The dataset contains 660 credit-card customers. Five behavioural attributes are used for segmentation; identifier columns are dropped before modelling.

Data dictionary

Variable	Description
Avg_Credit_Limit	Average credit limit of the customer (USD)
Total_Credit_Cards	Number of credit cards the customer holds
Total_visits_bank	Number of visits the customer made to a bank branch
Total_visits_online	Number of online visits / logins by the customer
Total_calls_made	Number of calls the customer made to the bank
Sl_No	Identifier, dropped before modelling
Customer Key	Identifier, dropped before modelling

3. Exploratory Data Analysis

3.1 Data overview & quality

Summary Statistics

	count	mean	std	min	25%	50%	75%	max
Sl_No	660.0	330.500000	190.669872	1.0	165.75	330.5	495.25	660.0
Customer Key	660.0	55141.443939	25627.772200	11265.0	33825.25	53874.5	77202.50	99843.0
Avg_Credit_Limit	660.0	34574.242424	37625.487804	3000.0	10000.00	18000.0	48000.00	200000.0
Total_Credit_Cards	660.0	4.706061	2.167835	1.0	3.00	5.0	6.00	10.0
Total_visits_bank	660.0	2.403030	1.631813	0.0	1.00	2.0	4.00	5.0
Total_visits_online	660.0	2.606061	2.935724	0.0	1.00	2.0	4.00	15.0
Total_calls_made	660.0	3.583333	2.865317	0.0	1.00	3.0	5.00	10.0

Overview

- The dataset contains 660 records and 7 columns, with no missing values.
- Each customer is identified by a unique customer key, which will not be considered for analysis, similar to the Sl_No.

NOTE:

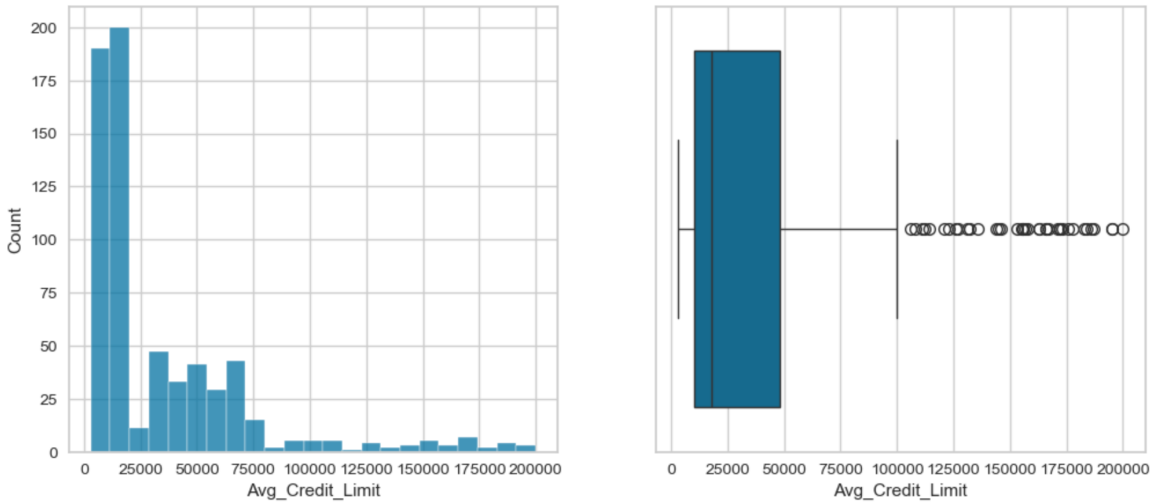
- The customer key has 5 duplicate values. Since the reason for these duplicates cannot be confirmed from the available data, they will not be treated as anomalies.

Numerical Features

- The average credit limit in this dataset ranges from USD 3,000 to USD 200,000. The currency is assumed to be denominated in USD. The median value is USD 18,000, indicating that the variable is heavily right-skewed.
- The total number of credit cards ranges from 1 to 10, with a median of 5.
- The total number of bank visits ranges from 0 to 5, with a median of 2.
- The total number of online visits ranges from 0 to 15, with a median of 2 and a mean of 2.6.
- The total number of calls made ranges from 0 to 10, with a median of 3.

3.2 Univariate analysis

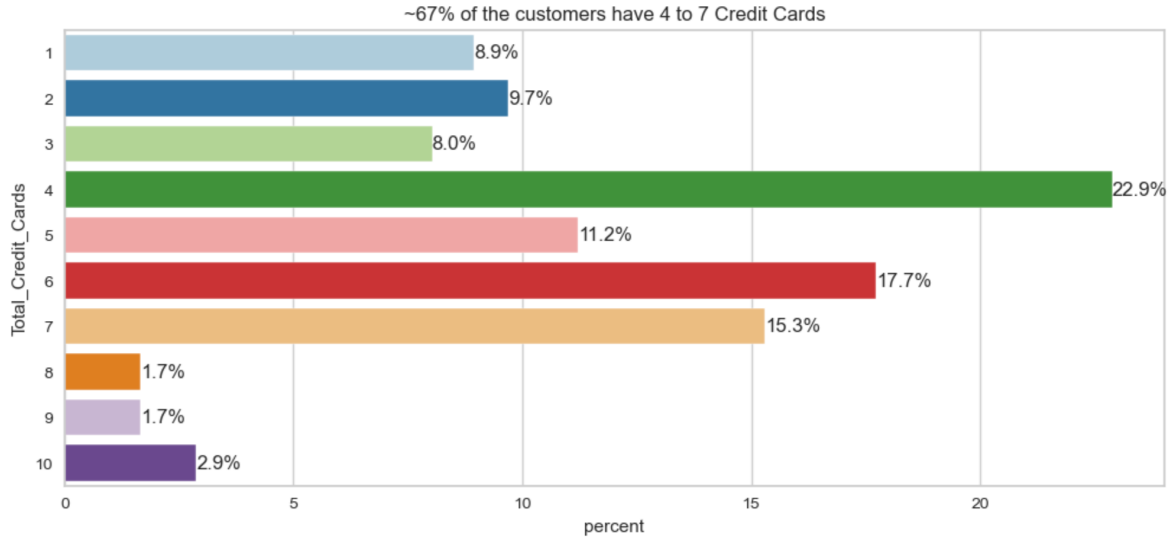
1. Average Credit Limit



Observation

The Average Credit limit distribution is heavily right skewed and multimodal indicating that there may be different customer groups characterised by their average credit limit. There are high value credit limits present in the upper tail.

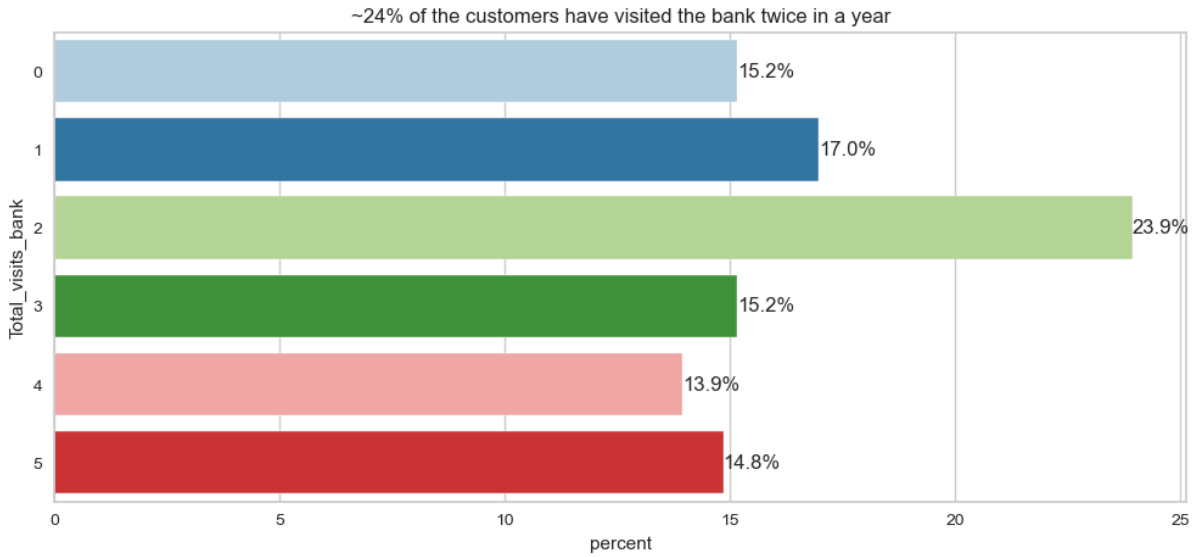
2. Total_Credit_Cards



Observation

~67% of the customers possess 4 to 7 credit cards. This group can be a potential upselling target.

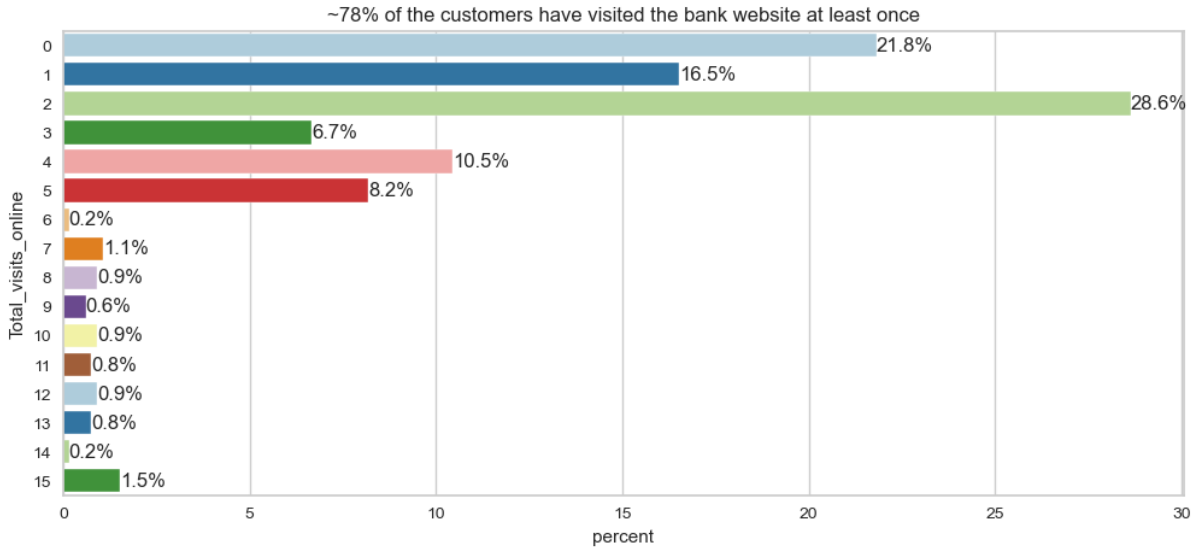
3. Total_visits_bank



Observation

15% of the customers have never visited the bank in a year, however need to check if this group pay online visits instead.

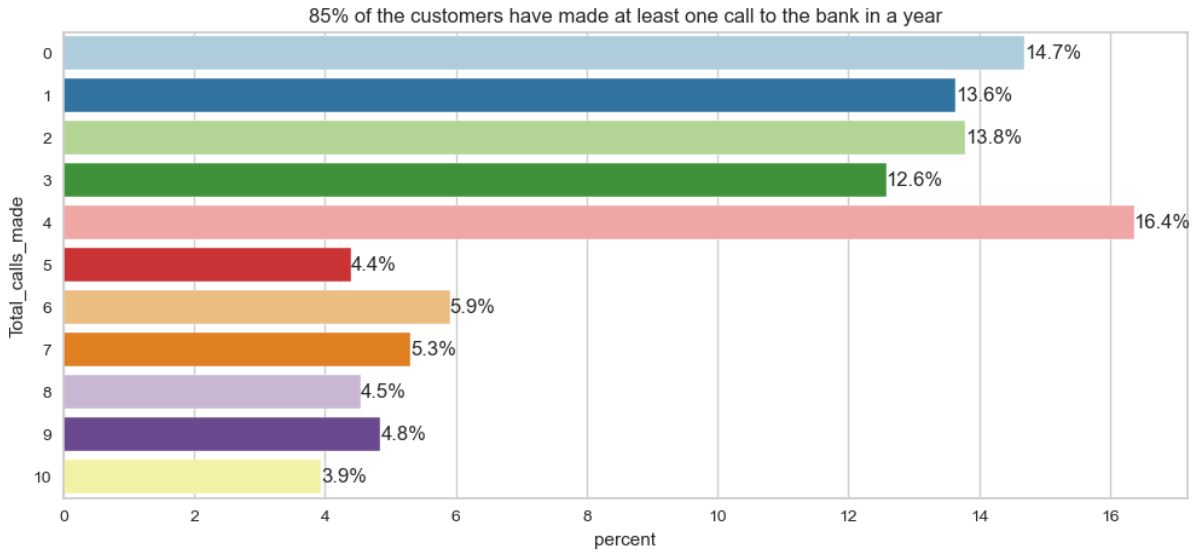
4. Total_visits_online



Observation

~78% of the customers have visited the bank website at least once in a one year period suggesting that digital adoption is prevalent.

5. Total_calls_made



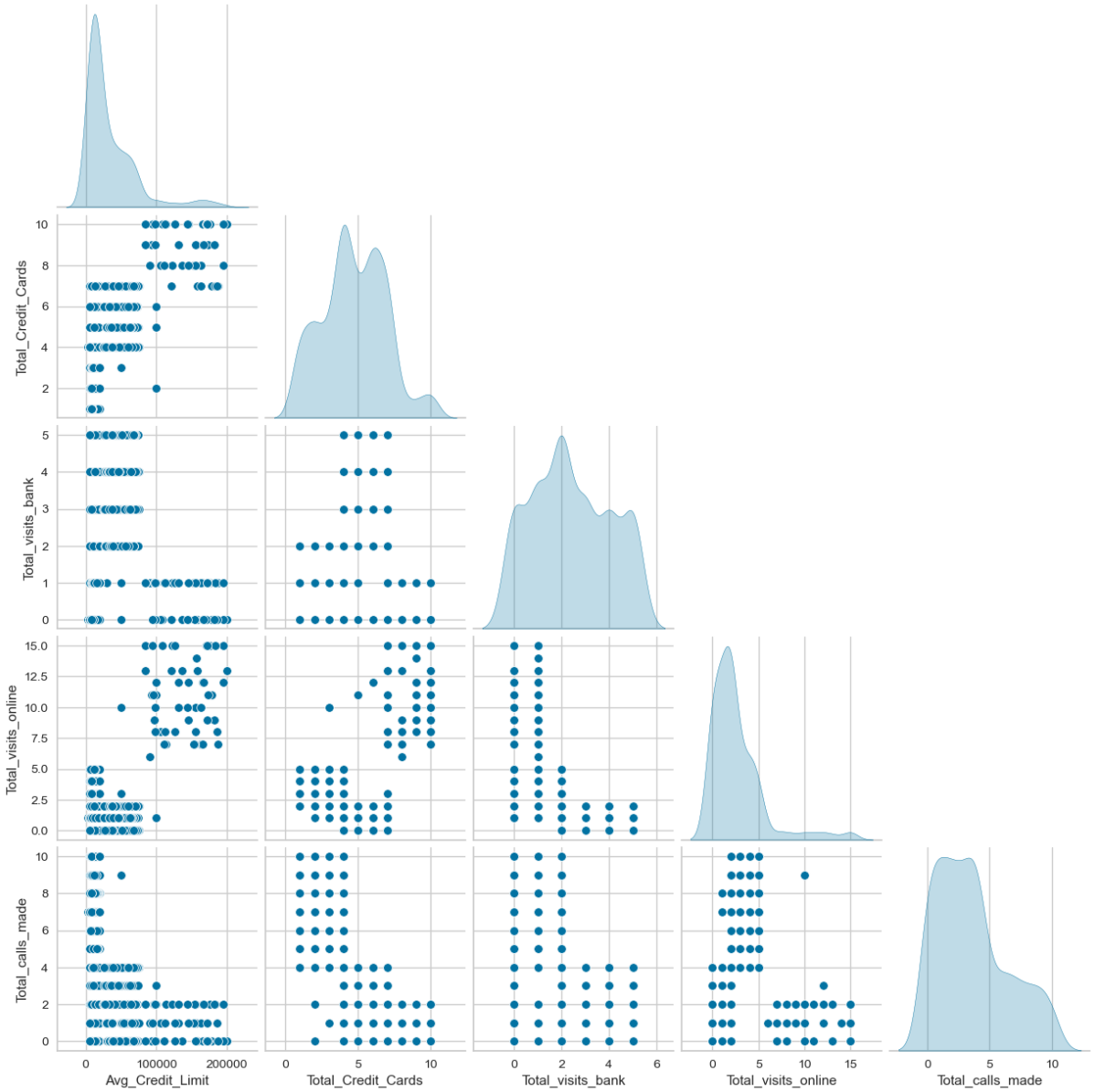
Observation

15% of the customers have not made any calls to the bank or the customer service indicating no phone based communication with the bank. This could reflect low engagement or possibly customers using other channels of communication.

~28% of the customers have made more than 4 calls to the bank in a year, indicating that this high engagement which may reflect service issue or assistance requirement.

3.3 Bivariate & correlation analysis

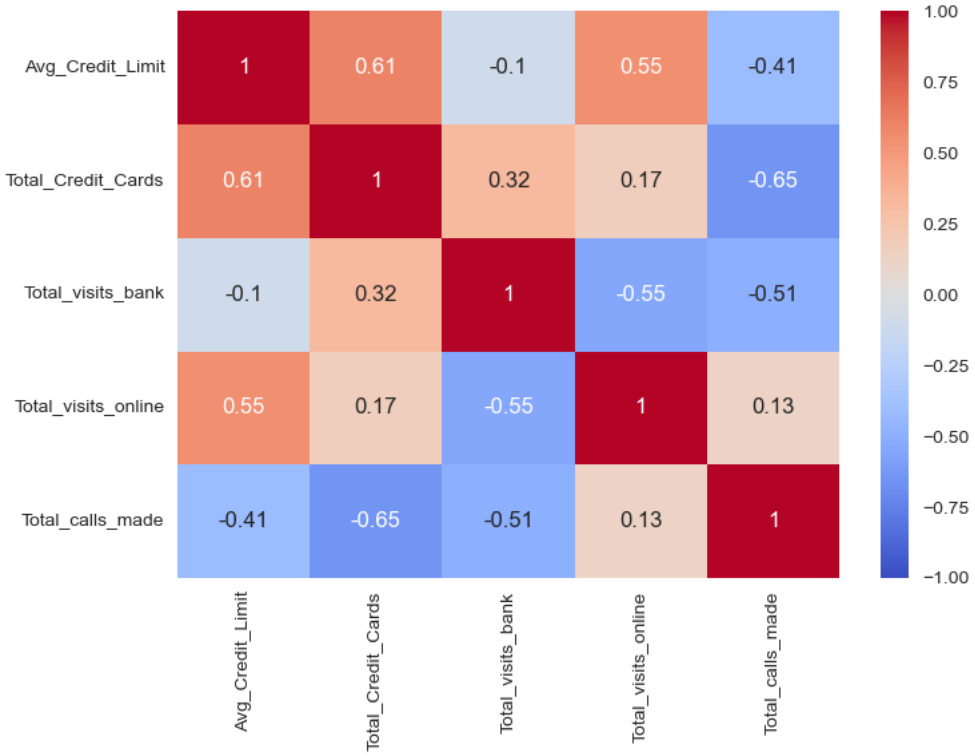
Pairplot of Numerical features



Observation

- The discrete bands in several plots are expected because cards, visits, and calls are count variables.
- The scatter patterns suggest the presence of distinct customer groups, especially a high-credit, high-online-usage group and a lower-credit, higher-call-usage group.

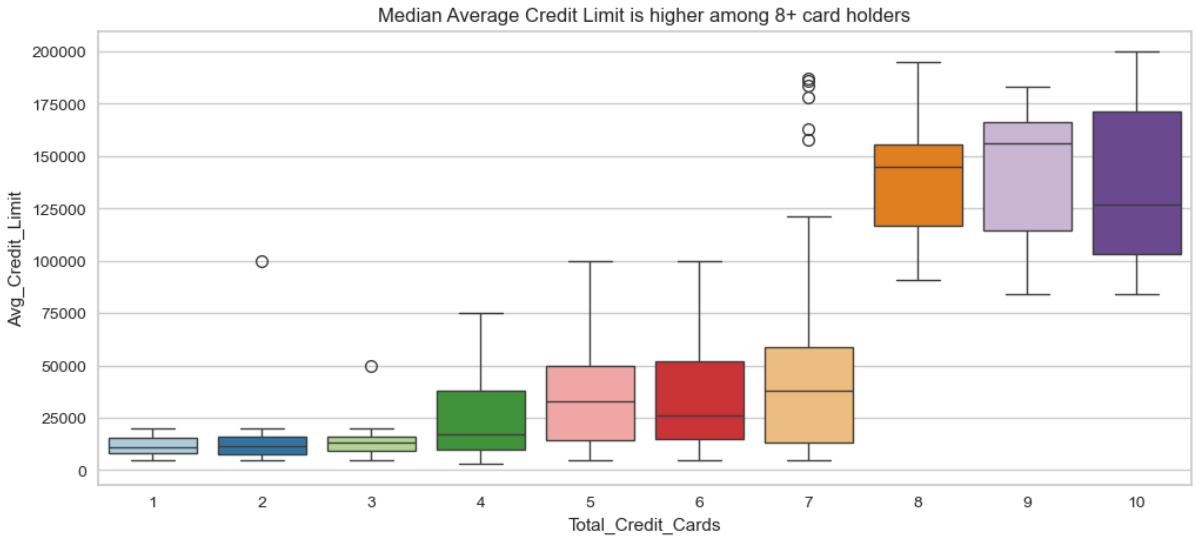
Correlation Analysis of Numerical Features



Observation

- Customers with higher credit limits appear to possess more credit cards and engage more through online channels. Customers with lower credit limits tend to engage more through phone calls and physical bank visits.
- `Total_Credit_Cards` and `Total_calls_made` show a strong negative correlation.
- Online visits and bank visits also exhibit a moderate negative correlation, suggesting different preferred modes of interaction.
- Average credit limit and bank visits have a weak negative correlation of **-0.1**.

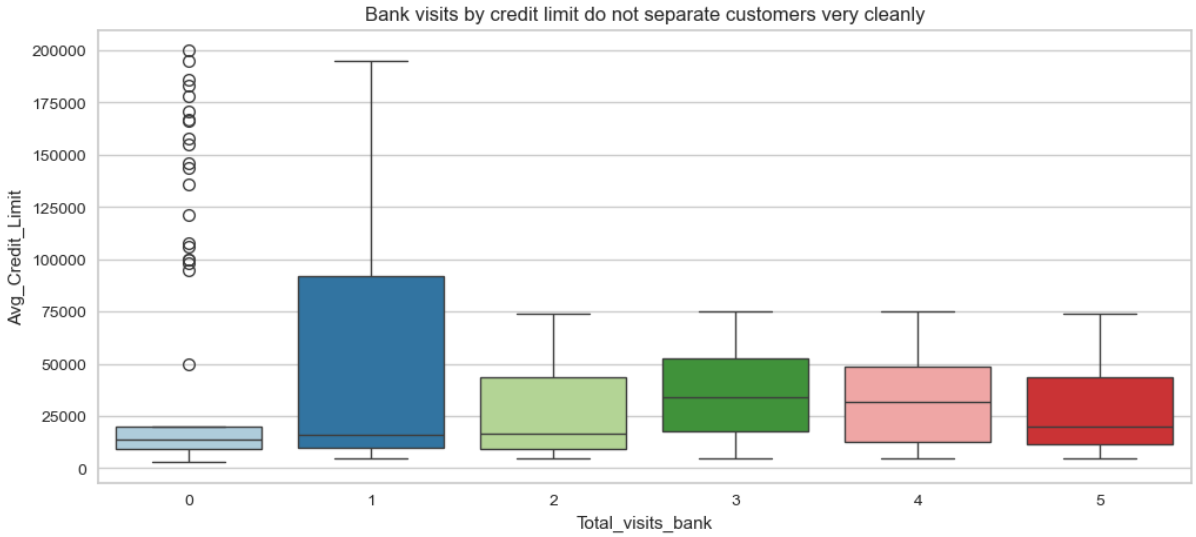
Avg_Credit_Limit by Total_Credit_Cards



Observation

- The Credit limit range of those with 8+ cards sits much higher than rest of the customers, forming a distinct group.
- Customers with 1-3 cards clearly have their average credit limit distribution almost identical and relatively lower compared to the rest of the customers, with very few outliers
- Customers with 4-7 cards show moderate but more variable limits

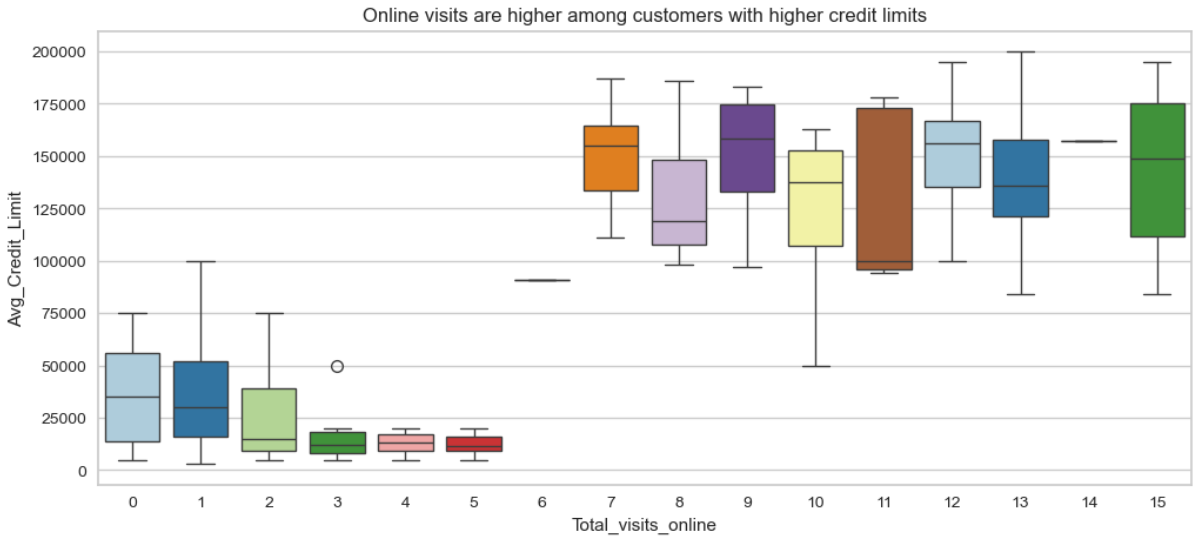
Avg_Credit_Limit by Total_visits_bank



Observation

There is no clear relationship observed in bank visits by credit limit which is already seen in the correlation matrix where Avg_Credit_Limit shows a weak negative correlation -0.1 which does not establish a strong linear relationship.

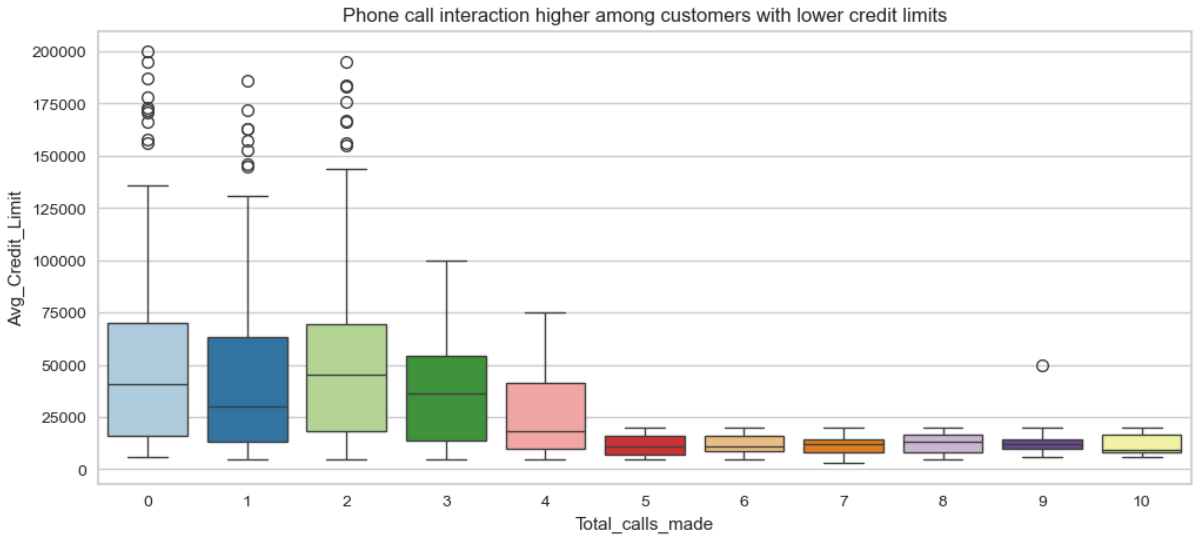
Avg_Credit_Limit by Total_visits_online



Observation

- Customers with 0-5 online visits are concentrated among lower avg credit limits
- Customers with 6-10 online visits are concentrated among higher avg credit limits with moderate variation

Avg_Credit_Limit by Total_calls_made



Observation

- Customers with higher credit limits appear to have made <5 phone calls to the bank in a year
- Customers with lower credit limits appear to have made 5+ calls to the bank in a year with their credit limit distribution substantially overlapping

3.4 Key takeaways from EDA

The analysis found that the Average credit limit distribution is heavily right-skewed with a multimodal pattern indicating that there may be multiple groups/segments of customers characterised by credit limit. This is associated to the possession of multiple credit cards suggesting that customers who own multiple credit cards have a higher credit limit compared to those who own fewer credit cards.

```
> The group that possesses 1–3 cards appear to have the lowest
Average credit limit

> The group with 4–7 cards have moderate credit limit with some
variation

> The group with 8+ cards have higher credit limits compared to
the rest of the customers
```

The bank visits in a year recorded for customer does not show any strong relationship with the Average credit limit whereas, customers who have higher average credit limit appear to engage more online and engage less through phone calls.

The customers with lower credit limit tend to possess fewer cards in general and appear to utilise phone based communication with the bank more than other forms of communication (online/physical).

The potential profiles suggested broadly by EDA are as follows:

```
> Higher credit limit –> more cards –> more online visits –> fewer
phone calls

> Lower credit limit –> fewer cards –> more phone calls
```

4. Data Preprocessing

4.1 Missing Value Treatment

The dataset contains no missing values; therefore, no imputation was required.

4.2 Outlier Detection and Treatment

- `Avg_Credit_Limit` was heavily right-skewed and contained genuine high-value observations.
- These observations were **not removed**, because they may represent a meaningful customer segment.
- Instead, `Avg_Credit_Limit` was log-transformed using `log1p()` to reduce skewness and limit the influence of very large values on distance-based clustering.
- Count variables such as cards, calls, and visits were checked for plausible ranges and retained because their extreme values were valid customer behaviour, not obvious data errors.

4.3 Feature Engineering

Two feature representations were evaluated.

Baseline pipeline

- `Avg_Credit_Limit`
- `Total_Credit_Cards`
- `Total_visits_bank`
- `Total_visits_online`
- `Total_calls_made`

No additional behavioural features were created.

Feature-engineered pipeline

In addition to Avg_Credit_Limit and Total_Credit_Cards, this pipeline created:

- `Total_interactions`
- `bank_share`
- `online_share`
- `calls_share`

Total_interactions = bank visits + online visits + calls

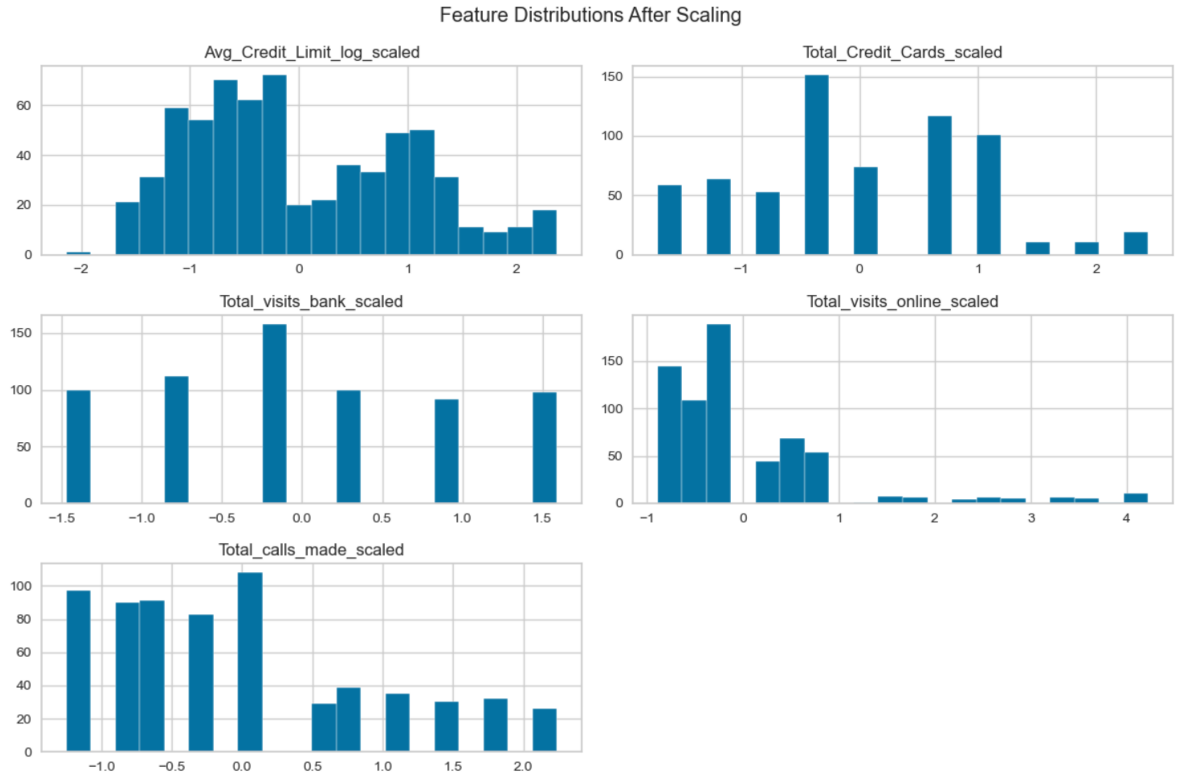
And each channel share as,

channel visits / Total_interactions

These features were created to capture both overall engagement intensity and preferred interaction channel, rather than relying only on raw counts.

4.4 Data Scaling

StandardScaler was applied because both KMeans and Hierarchical Clustering are distance-based algorithms. Without scaling, features with larger numerical ranges, particularly `Avg_Credit_Limit`, could dominate the distance calculations.



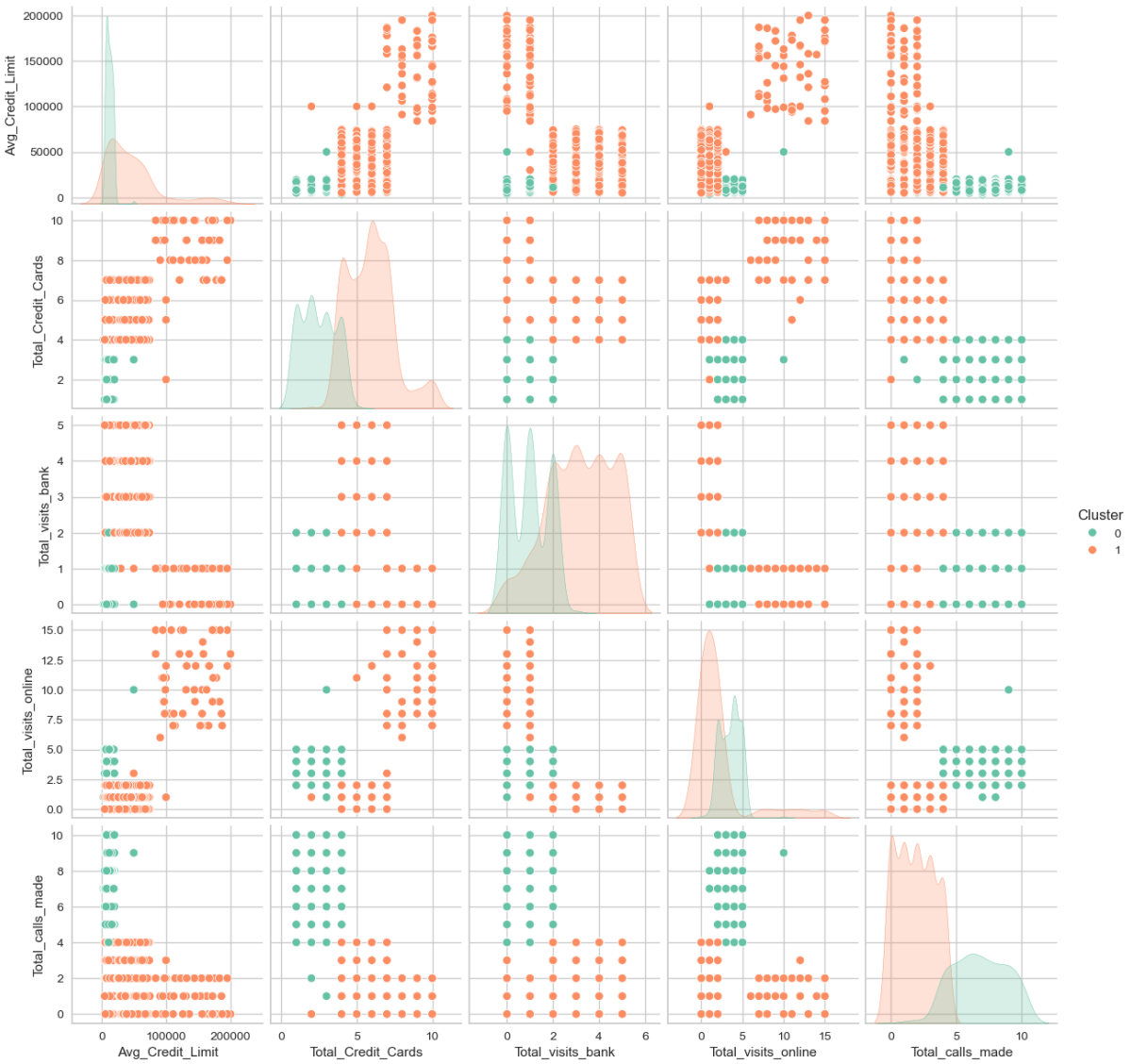
Observation

The transformed features have means approximately equal to 0 and standard deviations approximately equal to 1, confirming that standardization was applied correctly and that all variables contribute on a comparable scale to the distance-based clustering algorithms.

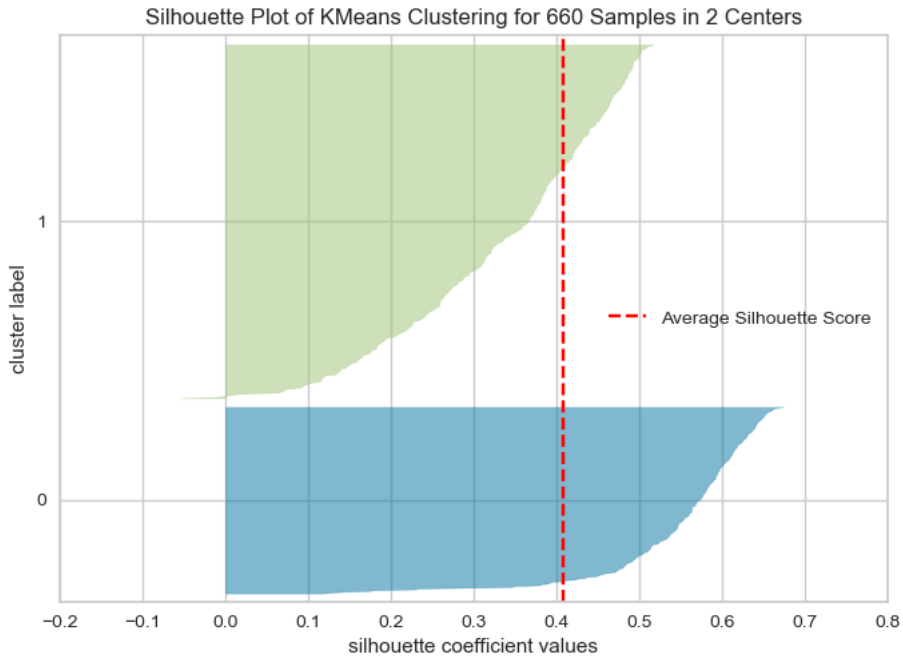
5. K-Means Clustering

5.1 Baseline K-Means

The EDA suggests two broad profiles (since the moderate credit limit group exhibits mixed behaviour), we will initialize a baseline clustering model with **k = 2 using original feature representation after preprocessing.**



The two-cluster solution separates a distinct high-credit-limit, high-card, high-online-usage, low-call customer group from the rest of the customer base. However, the second cluster contains customers with noticeably different interaction patterns, including branch-oriented and call-heavy customers. This suggests that `k = 2` may be too broad to capture the underlying customer segments effectively.



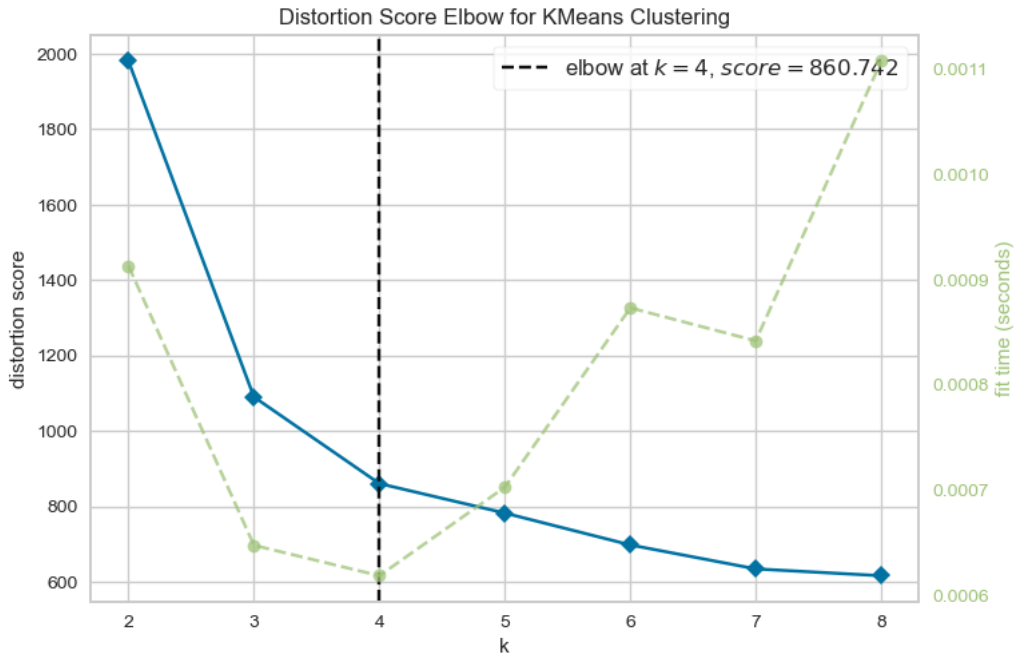
Observation

The 2 cluster baseline identifies the distinct lower credit limit, phone-oriented customer group. However, cluster 1 shows a wide spread in the customer behaviour. This suggests that k = 2, that is, 2 clusters maybe too few to capture the different customer profiles in the data.

This is further validated by the Silhouette Diagram where the Cluster 1 extends to the left of zero, the silhouette score approaching the negative region, suggesting that few instances in Cluster 1 may be poorly assigned.

Elbow Curve

Using the elbow method, we evaluated the distortion score across k in the range 2 to 8.

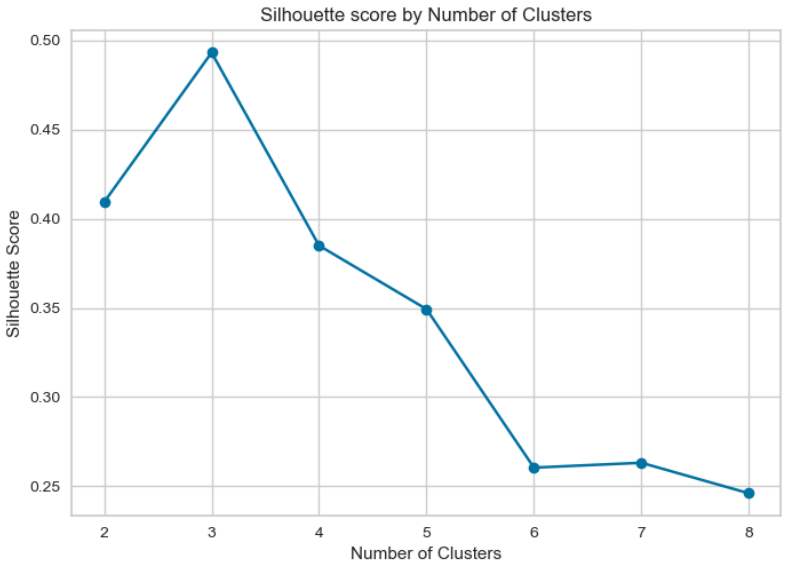


Observation

The inertia drops very quickly as we increase k up to 4 but then it decreases much more slowly as we keep increasing k. The drop from 2 to 3 is much larger than the drop from 3 to 4 so we shall consider k = 3 as the elbow.

So if we did not know better, **k = 3 will be a good choice** for the number of cluster.

Silhouette Analysis

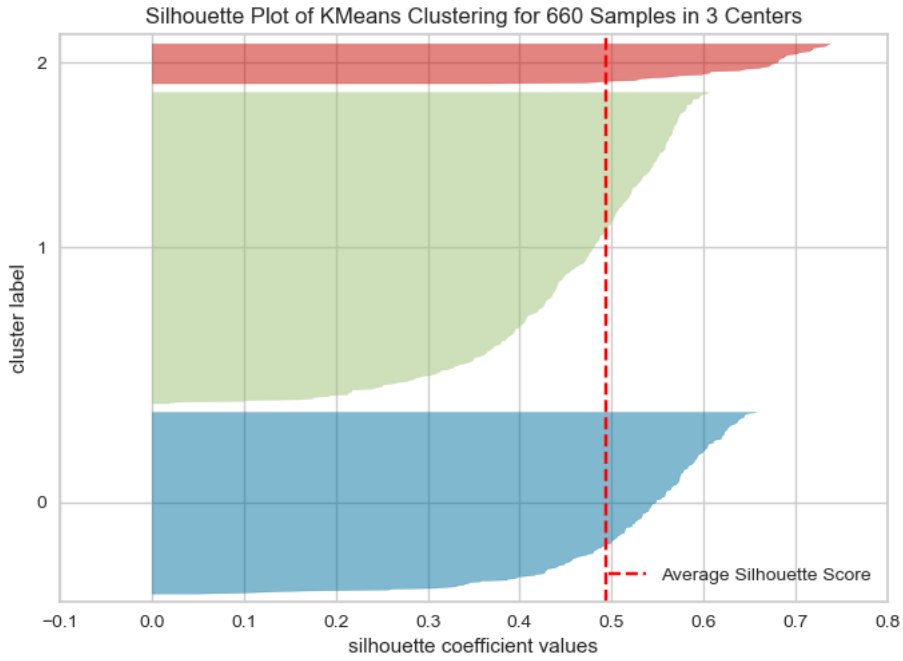


Observation

k	Inertia	Silhouette Score
2	1980.65	0.409
3	1090.34	**0.493**
4	860.74	0.385
5	775.81	0.349
6	697.64	0.261
7	634.21	0.263
8	598.80	0.246

This silhouette score analysis confirms that **k = 3 is a very good choice** as the **silhouette score is highest when k = 3.** A value closer to 1 means that the instance is well inside its own cluster and far from other clusters.

Silhouette Diagram



Observation

- All three clusters are entirely to the right of zero, a very few are near zero which represent the poorly assigned instances
- The vertical dashed line 0.49 is the average silhouette score and the knife shape extends beyond the dashed line towards 1
- The bands are reasonably thick, the thickness representing the number of instances. Cluster 2 appears to have the least number of instances, while cluster 1 has the most number of instances.

Selecting the Optimal number of Clusters

The appropriate number of clusters was determined using a combination of the elbow method, silhouette scores, and business interpretability.

- The inertia curve shows a sharp reduction from `k=2` to `k=3`, after which the improvement becomes more gradual, indicating an elbow around `k=3`.
- The silhouette score reaches its highest value at `k=3` with a score of **0.493**, compared with 0.409 at `k=2` and lower scores for larger values of `k`.
- From a business perspective, three clusters produce distinct and actionable customer segments, separating high-value, moderate-value and low-value customers.

Based on the elbow curve, silhouette score, and business interpretability, **k = 3** was selected as the optimal number of clusters for the KMeans model.

5.2 Feature Engineered and comparison with Baseline k = 3

A clustering model with engineered features was built to assess if the added complexity was justified. The following comparison table summarises how the feature engineered clustering model performed.

Criterion	Baseline KMeans	Feature-Engineered KMeans
Feature Representation	Original features, with log transformation of Avg_Credit_Limit and scaling	Engineered interaction features (Total_interactions and channel shares), along with credit limit and credit cards
Number of Clusters	3	3
Silhouette Score	**0.493**	**0.456**
Inertia	1090.34	1483.47
Inertia Comparison	Not directly comparable across different feature spaces	Not directly comparable across different feature spaces
High-Value Segment	Stable cluster of 50 customers	Similar high-value segment
Cluster Profiles	High-value digital, moderate branch-oriented, and low-value high-call customers	Similar three customer profiles
Interpretability	High; based directly on original business variables	High, but requires interpretation of engineered interaction and channel-share features
Model Complexity	Lower	Higher due to feature engineering
Overall Observation	Better cluster separation with simpler features	Lower silhouette despite added feature engineering

Takeaway

At `k = 3`, the feature-engineered model offers better interpretability. However, it has a few instances with negative silhouette coefficient values, indicating possible misassignment or overlap between clusters.

In the baseline model with `k = 3`, all instances have silhouette scores in the positive range, suggesting that the grouping has been reasonably well separated.

Since the two models operate in different feature spaces, their silhouette scores should be compared with some caution. However, the baseline model achieves a higher silhouette score of **0.493**, compared with **0.456** for the feature-engineered model, while also producing well-separated and interpretable clusters.

Therefore, the **baseline feature representation is retained as the better KMeans solution**, as feature engineering does not provide sufficient improvement in cluster quality to justify the added complexity.

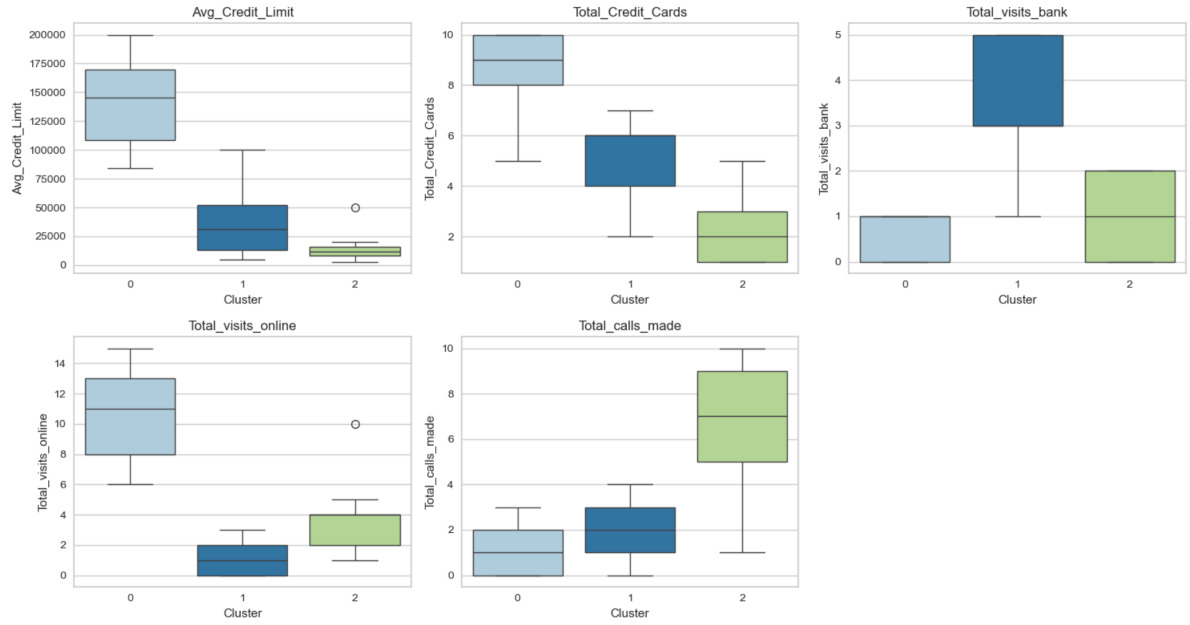
5.3 Cluster Profiling for Baseline KMeans model

The clustering model produces 3 distinct customer segments.



- Cluster 0: **50 customers**
- Cluster 1: **385 customers**
- Cluster 2: **225 customers**

Cluster 1 forms the largest customer segment, accounting for approximately 58% of the customer base, followed by Cluster 2 at around 34%. Cluster 0 is the smallest segment, representing about 8% of customers. Despite its smaller size, Cluster 0 remains a distinct and consistently identified high-value customer group.



- The boxplots show that the three clusters are differentiated not only by credit profile but also by preferred interaction channels.
- The high-value cluster is strongly digital, the moderate-value cluster is branch-oriented, and the low-value cluster is more call-dependent.

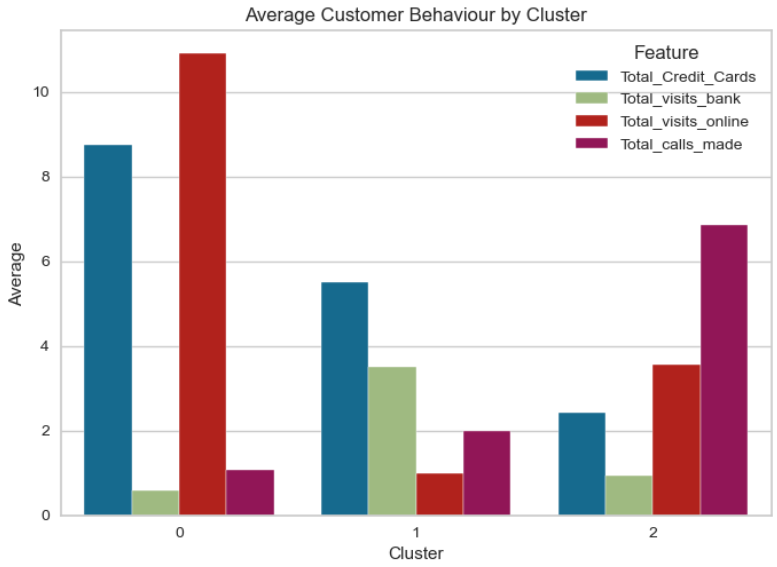
The mean of value of individual numerical features by cluster label is given below:

Cluster	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
0	141040.000000	8.740000	0.600000	10.900000	1.080000
1	33851.948052	5.516883	3.493506	0.979221	1.994805
2	12151.111111	2.422222	0.937778	3.546667	6.857778

Cluster	Segment Name	Customer Characteristics
0	**High-value digital-first customers**	High credit limits, more credit cards, very high online engagement, and minimal branch or phone interaction.
1	**Moderate-value branch-oriented customers**	Moderate credit limits and card ownership, with a clear preference for branch visits and relatively low online and phone engagement.
2	**Low-value call-heavy customers**	Lower credit limits, fewer credit cards, high call activity, and moderate online engagement.

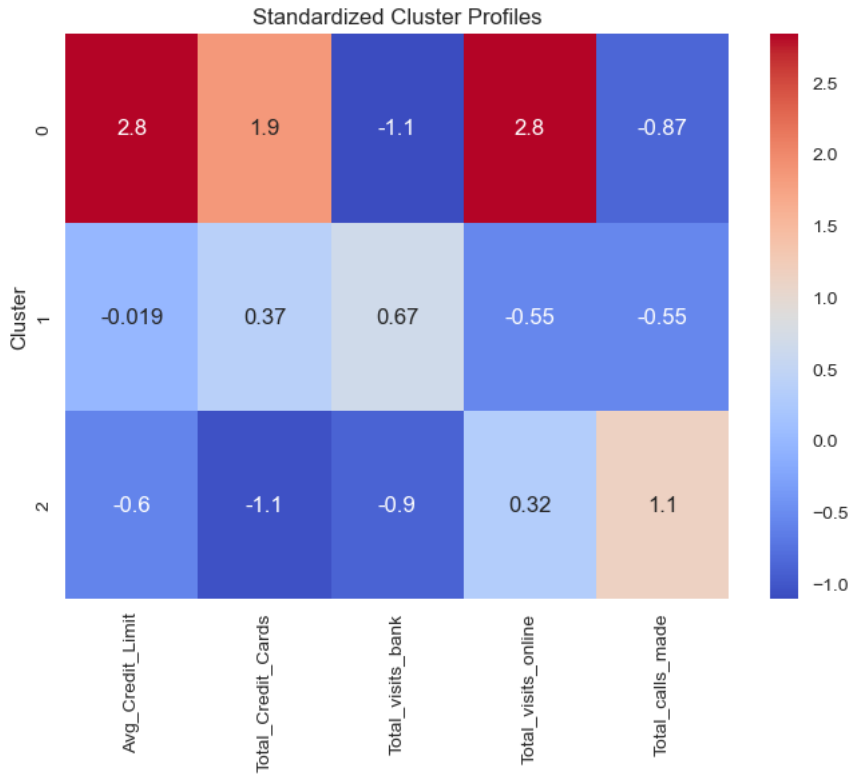
The following visualizations further illustrate the behavioural differences between the identified customer segments.

What does the average customer in each cluster actually do?



This chart shows how an average customer in each cluster behaves and how distinct they are from other clusters.

How Does Each Cluster Differ From The Average Customer?



The barplot and standardized heatmap confirms that these behavioural patterns are distinct relative to the overall customer base.

6. Hierarchical Clustering

6.1 Baseline Hierarchical Clustering Model

Approach

Hierarchical Clustering was applied to the processed baseline dataset, comparing multiple linkage methods and distance metrics such as,

Linkage Method	Distance Metric
Single	Euclidean
Single	Cityblock
Complete	Euclidean
Complete	Cityblock
Average	Euclidean
Average	Cityblock
Weighted	Euclidean
Weighted	Cityblock
Ward	Euclidean
Centroid	Euclidean

The model selection was done based on Dendrogram clarity, Cophenetic Correlation, Silhouette Score and business interpretability.

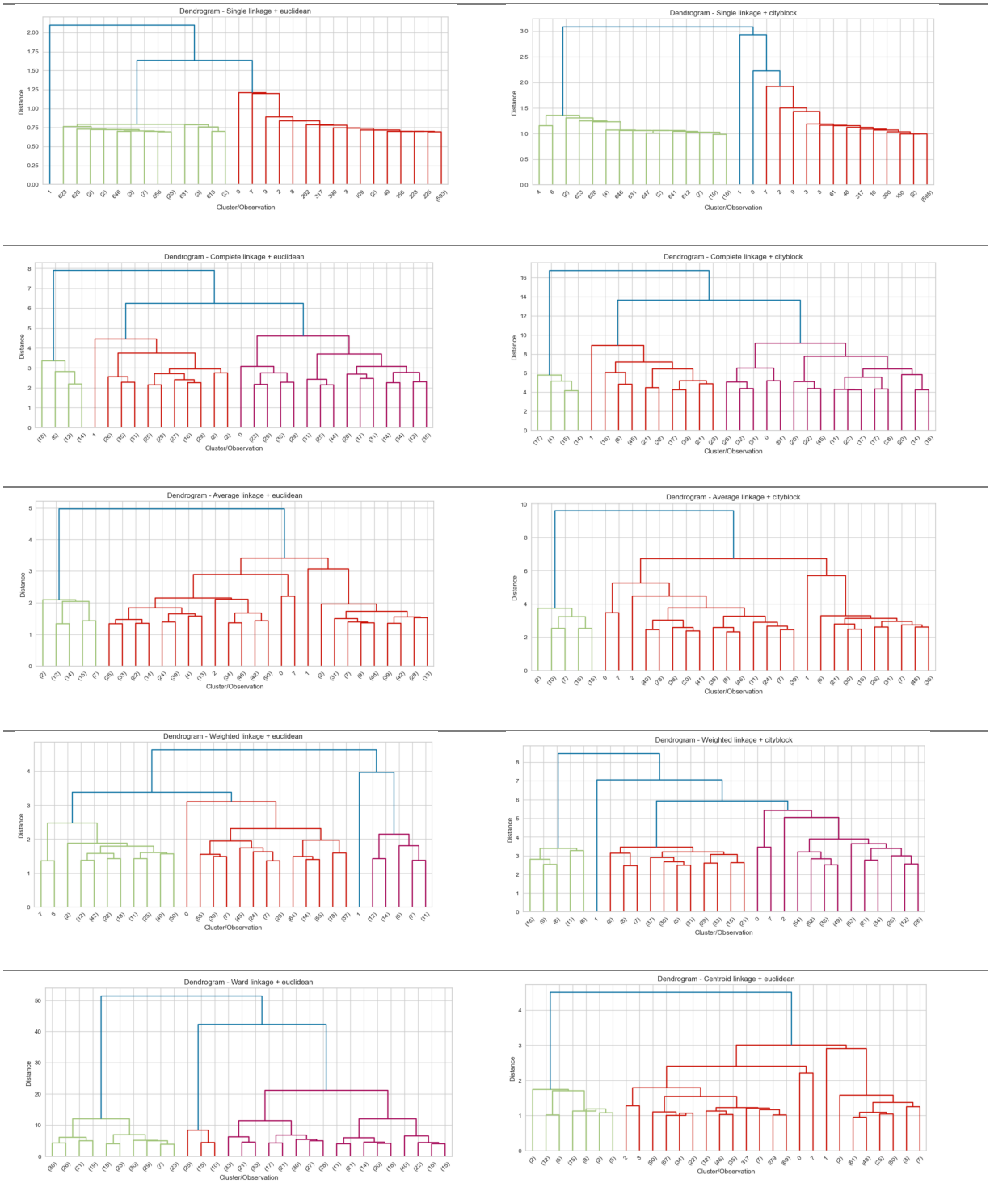
6.2 Comparing Linkage Methods

Cophenetic Correlation table

The Cophenetic Correlation found for the linkage methods are represented by the table below.

Rank	Linkage Method	Distance Metric	Cophenetic Correlation
1	Average	Euclidean	0.8693
2	Centroid	Euclidean	0.8684
3	Weighted	Euclidean	0.8627
4	Average	Cityblock	0.8609
5	Weighted	Cityblock	0.8539
6	Complete	Euclidean	0.8381
7	Complete	Cityblock	0.8319
8	Ward	Euclidean	0.7465
9	Single	Euclidean	0.6936
10	Single	Cityblock	0.6727

Dendrograms For Various Linkage Methods x Distance Metrics



Average Linkage and Ward Linkage appear to have shorter merges compared to the other combinations of Linkage and Distance Metrics. Centroid Linkage exhibits non-monotonic merges which makes business interpretability more complex.

Candidate Linkage Methods and Distance Metrics

Linkage + Distance	Reason for Selection
Average + Euclidean	Selected due to its high cophenetic correlation score and clear dendrogram structure.
Average + Cityblock	Selected due to its strong cophenetic correlation score and clear dendrogram structure.
Ward + Euclidean	Included as a robustness check to compare whether a compact, variance-minimizing linkage produced a similar cluster structure.

Centroid linkage was excluded from further consideration because its dendrogram showed non-monotonic merges, making the hierarchy less straightforward to interpret.

6.3 Selecting the Best Linkage Method, Distance Metric and Number of Clusters

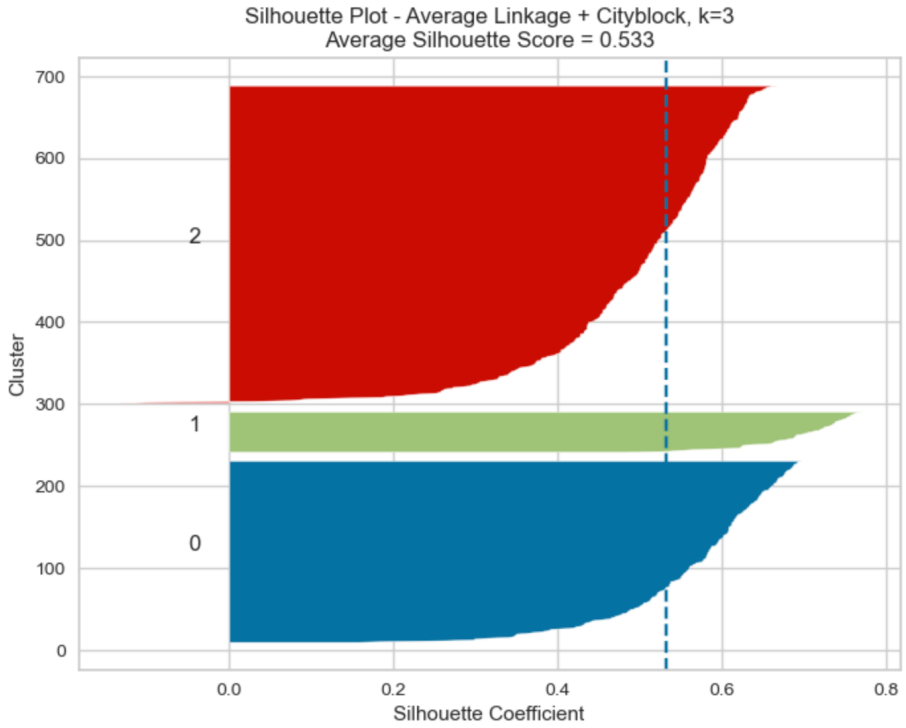
Three criteria were used to select the final hierarchical clustering configuration:

1. **Cophenetic correlation (hierarchy fidelity)** - Average linkage with Euclidean distance ranked highest (0.8693), closely followed by Average linkage with Manhattan distance (0.8609). Both exceed the 0.85 threshold typically used to indicate a trustworthy hierarchy.
2. **Silhouette score at the chosen k (cluster separation)** - Average linkage with Manhattan distance at **k=3** produced the highest silhouette score (0.533), outperforming Euclidean-based alternatives at the same k.

Linkage + Distance	No. of Clusters (k)	Silhouette Score
Average + Euclidean	2	**0.501**
Average + Euclidean	3	0.490
Average + Cityblock	**3**	**0.533**
Ward + Euclidean	3	0.493

   Although Average + Euclidean achieved a slightly higher silhouette score at `k=2` than at `k=3`, the three-cluster solution was preferred for its greater business interpretability. Among the three-cluster models, **Average + Cityblock produced the highest silhouette score of 0.533**, outperforming Average + Euclidean (0.490) and Ward + Euclidean (0.493). Combined with its strong cophenetic correlation (0.861) and clear dendrogram structure, Average + Cityblock appears to be the best candidate over all.

3. **Robustness considerations** - Manhattan distance is less sensitive to outliers than Euclidean, which is particularly relevant given the right-skewed distribution of Avg_Credit_Limit observed during EDA.



The overall silhouette score of 0.533 indicates reasonably good cluster separation. A small number of observations in one cluster have negative silhouette values, suggesting some overlap or possible misassignment near the cluster boundary. However, the majority of observations have positive silhouette coefficients, indicating that the three-cluster solution is generally well formed.

Given the small cophenetic correlation gap (0.0084) and the clear advantage in silhouette score and outlier robustness, Average linkage with Manhattan distance was selected as the final configuration for baseline model, **with k=3 clusters**.

6.4 Feature Engineered Hierarchical Model

To evaluate whether engineered features capturing overall interaction intensity and channel preference improve the hierarchical clustering solution, the same clustering workflow was repeated using the feature-engineered dataset. The resulting model was compared with the baseline hierarchical model based on silhouette score, cophenetic correlation, cluster structure, and interpretability.

Criterion	Baseline Hierarchical Model	Feature-Engineered Hierarchical Model
Feature Representation	Original features, with log transformation of Avg_Credit_Limit and scaling	Engineered interaction features (Total_interactions and channel shares), along with credit limit and credit cards
Linkage Method	Average	Average
Distance Metric	Cityblock	Cityblock
Number of Clusters	3	3
Silhouette Score	**0.533**	**0.492**
Cophenetic Correlation	**0.86**	**0.829**
Dendrogram	Clear 3-cluster structure	Clear 3-cluster structure
High-Value Segment	Stable cluster of 50 customers	Stable cluster of 50 customers
Cluster Profiles	High-value digital, moderate branch-oriented, and low-value high-call customers	Similar three customer profiles
Interpretability	High; clusters are directly explained using original business variables	High, but requires interpretation of engineered interaction and channel-share features
Model Complexity	Lower	Higher due to additional feature engineering
Overall Observation	Stronger cluster separation with simpler features	Does not improve silhouette despite additional feature engineering

The feature-engineered model identified broadly similar customer segments, but the silhouette score decreased from **0.533 to 0.492**. Since the engineered features did not improve cluster separation or produce meaningfully different business segments, **the baseline hierarchical model is the better hierarchical model solution.**

6.5 Cluster Profiling for Baseline Hierarchical Model



Cluster	Number of Customers	Percentage
0	222	33.6%
1	50	7.6%
2	388	58.8%
Total	**660**	**100%**

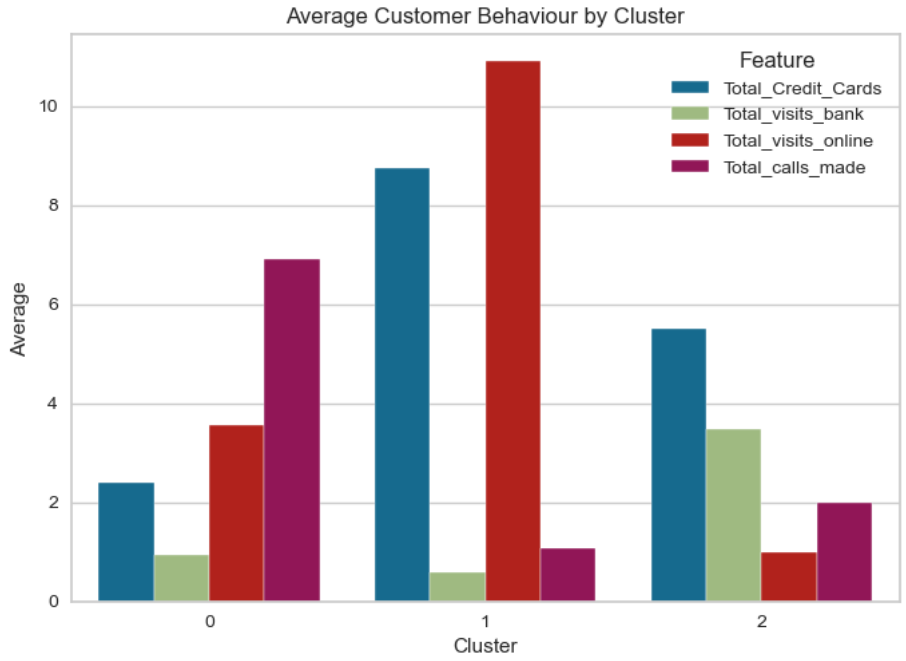
Profile Mean

Cluster	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
0	12184.684685	2.400901	0.932432	3.572072	6.909910
1	141040.000000	8.740000	0.600000	10.900000	1.080000
2	33664.948454	5.505155	3.476804	0.984536	2.002577

Qualitative customer profile

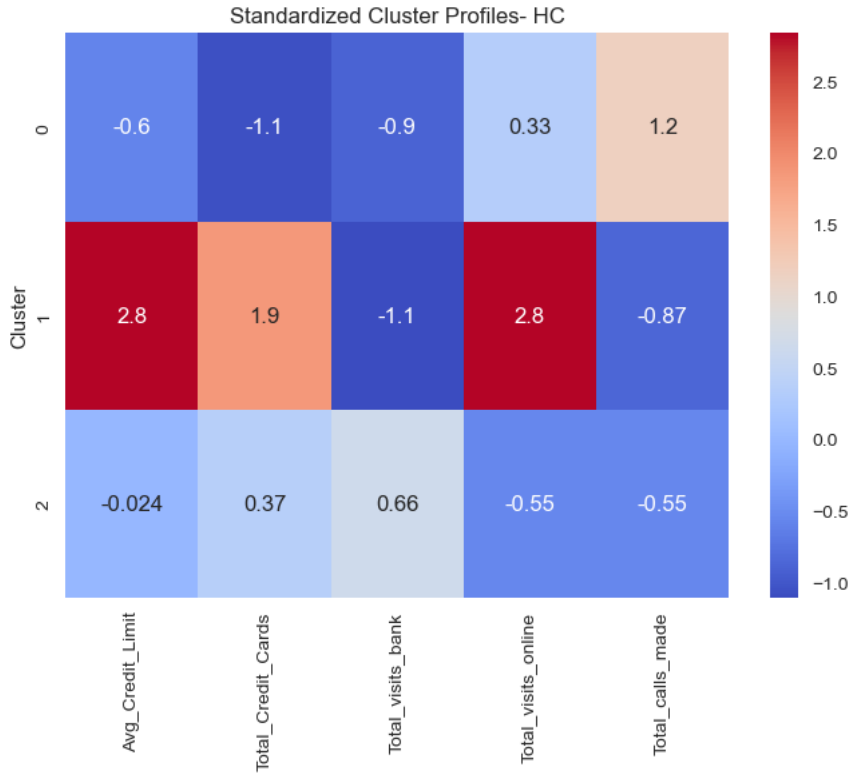
Cluster	Customer Profile	Credit Limit	Credit Cards	Bank Visits	Online Visits	Calls Made
0	**Low-value, call-heavy customers**	Low	Low	Low	Moderate	High
1	**High-value, digital-first customers**	Very High	High	Low	Very High	Low
2	**Moderate-value, branch-oriented customers**	Moderate	Moderate	High	Low	Low to Moderate

What Does An Average Customer Within A Cluster Do ?



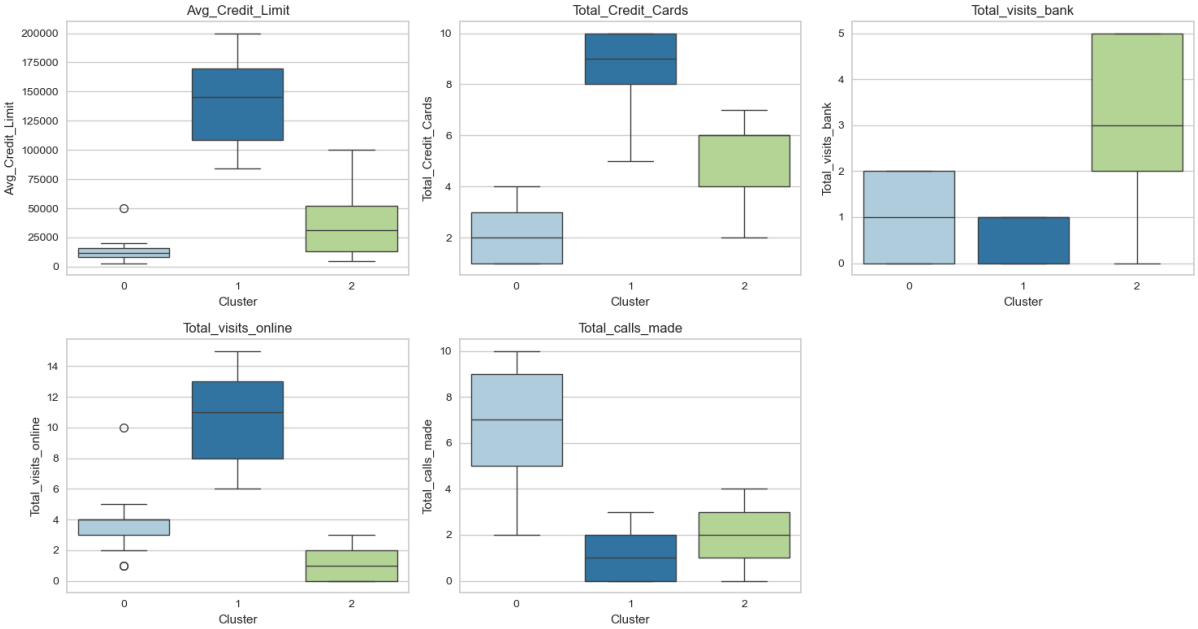
The chart shows three clearly different behaviour patterns. **Cluster 0** has fewer cards and bank visits but very high call activity, indicating a call-heavy segment. **Cluster 1** has the highest card ownership and online usage with very low branch and call activity, making it strongly digital-first. **Cluster 2** shows moderate card ownership, the highest branch usage, and low online activity, indicating a branch-oriented segment.

How Does Each Cluster Differ From The Average Customer ?



The heatmap confirms that the three clusters differ clearly from the overall customer average. **Cluster 1** is the most distinct, with much higher credit limits, card ownership, and online activity, but lower branch visits and calls. **Cluster 0** shows the opposite pattern, with lower credit limits and card ownership but much higher call activity. **Cluster 2** is closest to the average customer, but with higher branch usage and lower online and call activity.

Distribution of Features by Cluster labels



- The boxplots show that the three clusters are differentiated not only by credit profile but also by preferred interaction channels.
- The high-value cluster(1) is strongly digital, the moderate-value cluster (2) is branch-oriented, and the low-value(0) cluster is more call-dependent.

7. Comparison of Methods

The following comparison table shows how the two approaches compare and which solution is the best way forward.

Criterion	KMeans	Hierarchical Clustering
Final Model	KMeans, k=3	Average Linkage + Cityblock, k=3
Feature Set	Baseline features	Baseline features
Number of Clusters	3	3
Silhouette Score	0.493	**0.533**
High-Value Segment	50 customers	50 customers
Main Cluster Profiles	High-value digital, moderate branch-oriented, low-value high-call	Same three broad profiles
Cluster Separation	Good	**Better overall separation**
Cluster Stability	Good	Very good; high-value segment remains clearly isolated
Interpretability	High	High
Computation Time (Wall Time)	**1.68 ms**	100 ms
Computational Efficiency	Faster and more scalable	Slower due to hierarchical distance calculations
Best Use in This Project	Strong, efficient segmentation	Best overall clustering quality for this dataset

Both KMeans and Hierarchical Clustering identified the same three broad customer segments. However, Hierarchical Clustering with Average linkage and Manhattan distance achieved a higher silhouette score of 0.533 compared with 0.493 for KMeans, indicating better cluster separation. KMeans was computationally faster, but the runtime difference was negligible for this small dataset.

For this relatively small dataset, hierarchical clustering is feasible and gives slightly better cluster separation. However, for large-scale banking datasets with millions of customers, **KMeans would generally be preferred because it is significantly more computationally efficient and scalable.**

Cross Method Segment Agreement

KMeans Segment	HC: Low-value, Call-heavy	HC: High-value, Digital-first	HC: Moderate-value, Branch-oriented
High-value, Digital-first	0	**50**	0
Moderate-value, Branch-oriented	0	0	**385**
Low-value, Call-heavy	**222**	0	3

The two clustering approaches show very high agreement, with **657 of 660 customers (99.5%)** assigned to equivalent business segments. Only three customers were classified differently between the moderate-value and low-value segments.

Cluster Stability – Customer Migration Between Segments

As noted in the EDA, five Customer Key values appeared more than once in the dataset. Rather than treating these as data errors, these duplicates were retained on the hypothesis that they represent genuine customer snapshots captured at different points in time, reflecting changes in the customer's financial profile or interaction behaviour.

To test this hypothesis, the cluster assignments for the duplicate records were examined:

Customer Key	Record 1 (Segment)	Record 2 (Segment)	Movement
37252	Call-heavy	Branch-oriented	✓ Migrated
47437	High-value digital	Branch-oriented	✓ Migrated
50706	Branch-oriented	Branch-oriented	— Stable
96929	Branch-oriented	Branch-oriented	— Stable
97935	Call-heavy	High-value digital	✓ Migrated

NOTE: This behaviour is consistent across both methods.

3 out of 5 duplicate Customer Keys (60%) were assigned to different segments across their two records. If these duplicate records represent observations from different points in time, this suggests that customer segment membership may change over time.

- **Customer 47437** appears in both the high-value digital and moderate branch-oriented segments across its two records. **If these records represent different points in time**, this could indicate a downward shift in value and digital engagement and may warrant further investigation.

- **Customer 97935** appears in both the call-heavy and high-value digital segments. **If the duplicate records are longitudinal snapshots**, this could represent an upward shift toward higher value and stronger digital engagement.

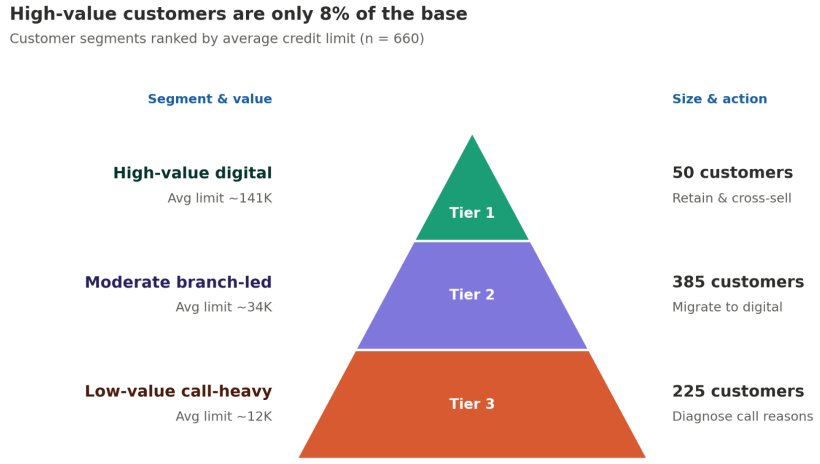
- **Customers 50706 and 96929** were assigned to the moderate branch-oriented segment in both records, indicating consistent segmentation across their duplicate observations. Given the

very small number of duplicate customers, this is **not sufficient to conclude that this segment is inherently more stable**.

Business Implications

- **Customer segmentation should be treated as potentially dynamic rather than permanently fixed.** If the duplicate records represent longitudinal observations, they suggest that customers may move between segments over time.
- The bank may have an opportunity to influence such movement through targeted engagement, for example by encouraging greater digital adoption among call-heavy customers. **This would need to be validated using longitudinal customer data.**
- In a production setting, periodically refreshing customer assignments, for example quarterly or semi-annually, would help ensure that targeting reflects current customer behaviour.

8. Insights & Recommendations



Recommendations are based on Baseline KMeans solution with k = 3 Clusters.

High-value, Digital first Customers

Key Insight : Very high credit limits(~141k), many cards, very high online usage, very few calls and branch visits

Marketing Recommendation: Target with premium cards, travel/reward offers, high-value cross-sell, personalized digital campaigns

Service Recommendation: Prioritize seamless app/web experience, proactive digital service, premium digital support

Moderate-value, branch-oriented customers

Key Insight : Moderate credit limits(~34k), more cards, higher branch usage, lower online usage

Marketing Recommendation: Promote suitable credit products and encourage gradual digital adoption through targeted offers

Service Recommendation: Maintain strong branch support while offering assisted digital onboarding and hybrid service options

Low-value, call-heavy customers

Key Insight : Lower credit limits(~12k), fewer cards, high call frequency, lower branch usage

Marketing Recommendation: Avoid aggressive premium-product targeting; focus on simple, relevant products and financial education

Service Recommendation: Improve call-center support, identify frequent contact reasons, offer self-service options for common issues